

Type of Graphical Analysis

- ↳ ① Univariate (only one column)
- ② Bivariate (Two columns / Features)
- ③ Multivariate. (More than Two)

Categorical / Numerical

① Univariate Analysis (Categorical feature)

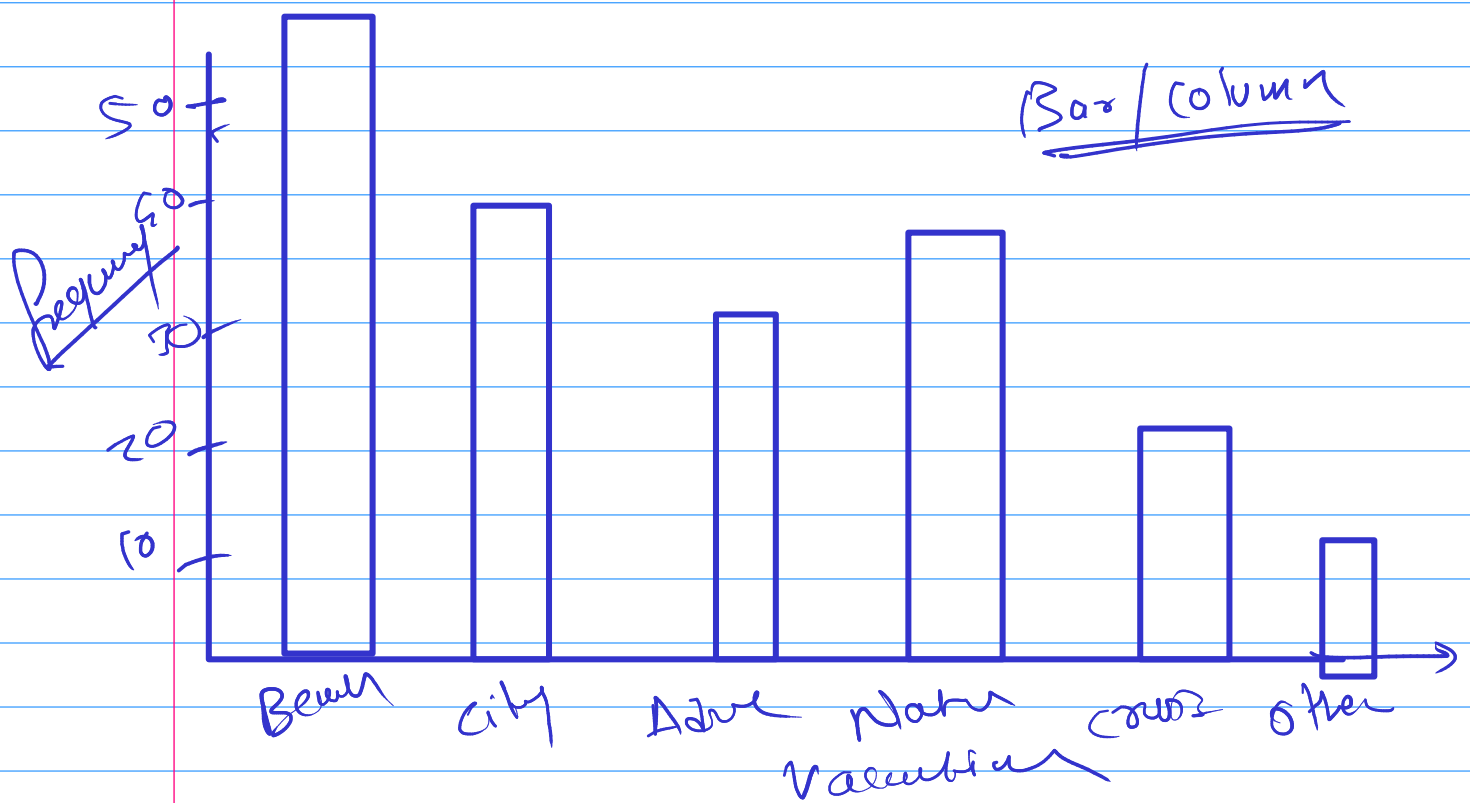
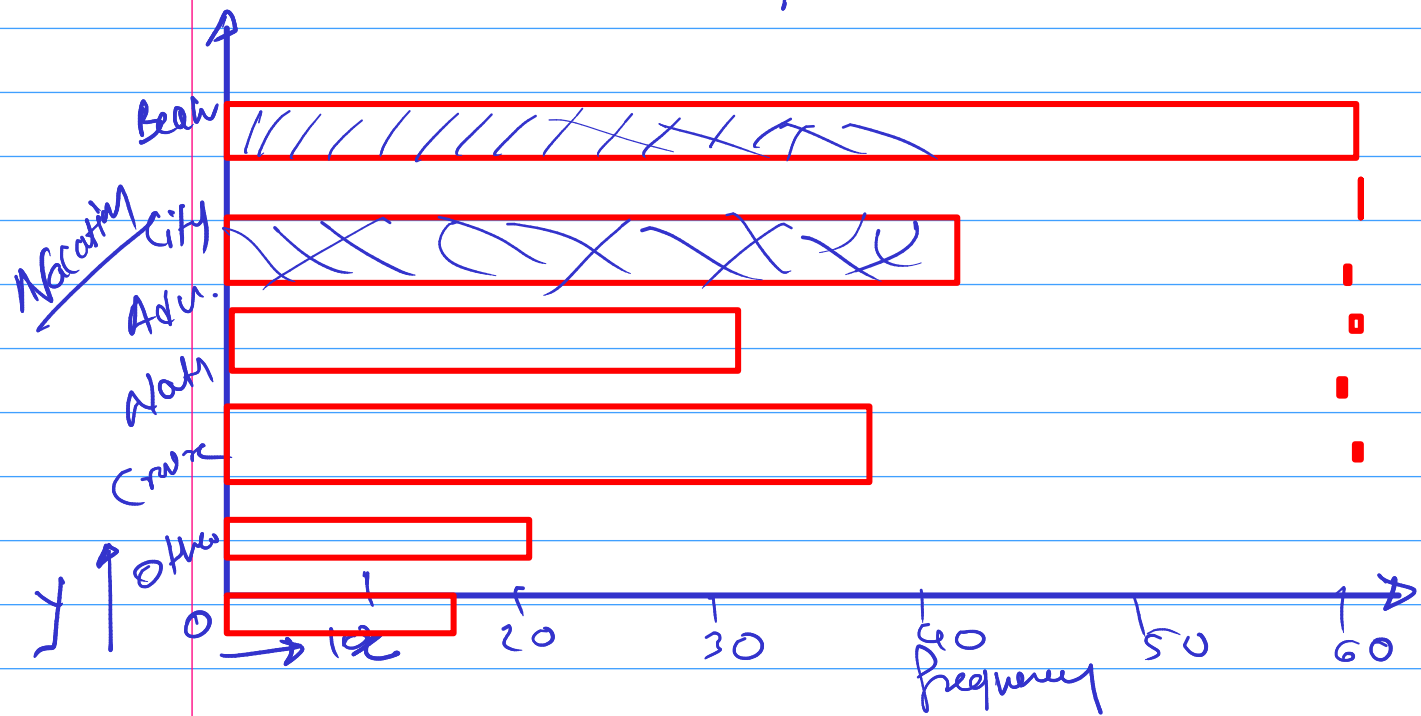
Picnic Data → Vacation → Categorical data
for 200 people.

Frequency Distribution Table

<u>Vacations</u>	<u>frequency (count)</u>
Beach	60
City	40
Adventure	30
Nature	35
Cruise	20
Other	15

Gender	freq
Male	03
female	02

① Bar chart / Column Chart.



Ratio / Percentage

Relative frequency

Valar

frequency

Relative freq (Ratio / percent)

Beach 60 $(60/200) \times 100 = 30\%$

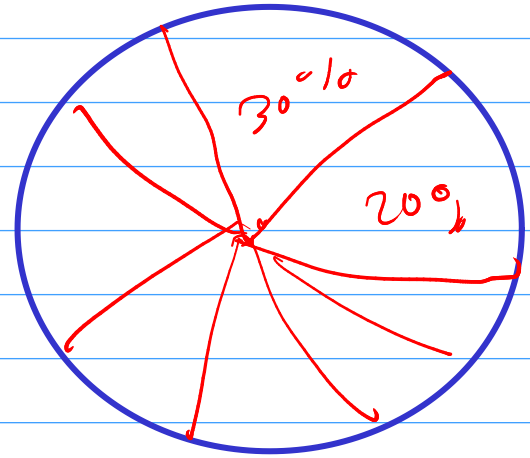
City 40 $(40/200) \times 100 = 20\%$

Advent 30 $= 15\%$

Nature 35 $= 17.5\%$

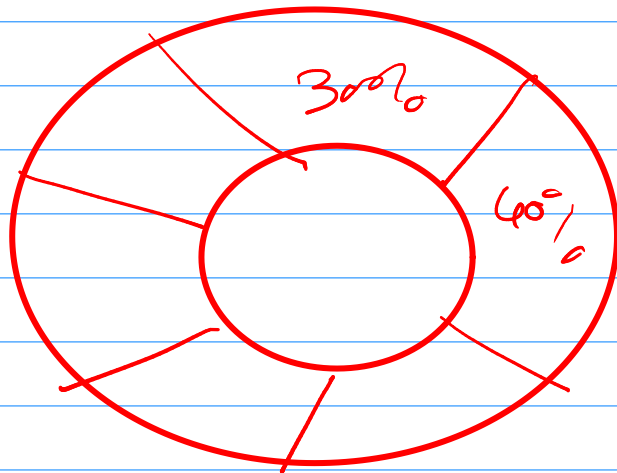
Course 20 $= 10\%$

Other 15 $= 7.5\%$



Pie-chart

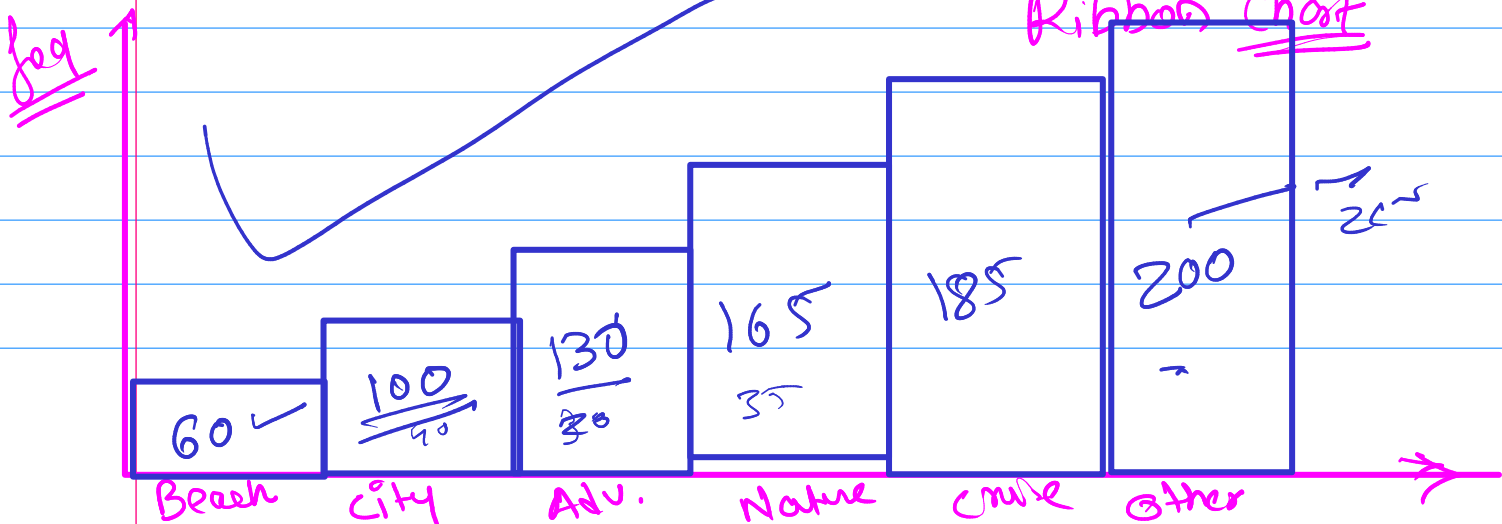
Donut Chart



Cumulative Frequency $\rightarrow$ addition

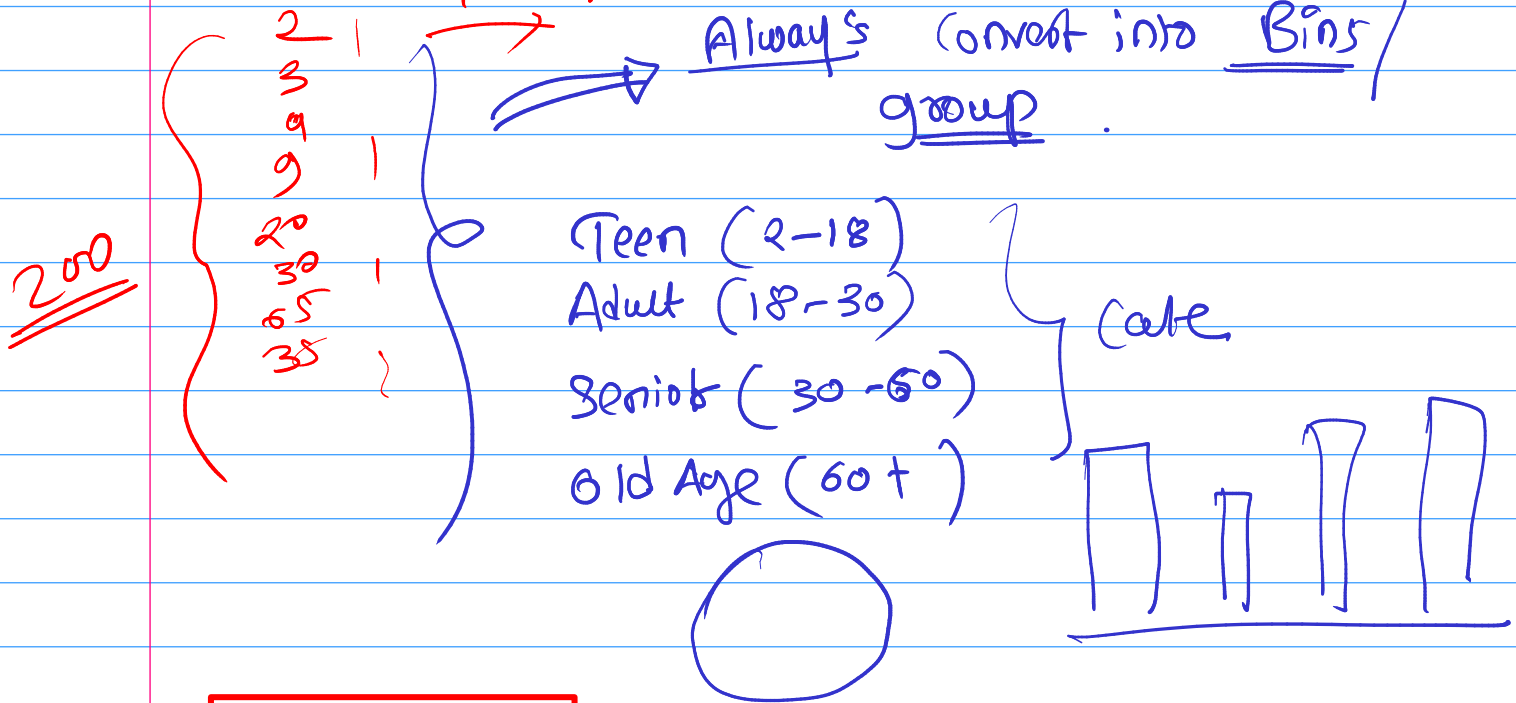
$\rightarrow$ waterfall chart

Ribbon chart

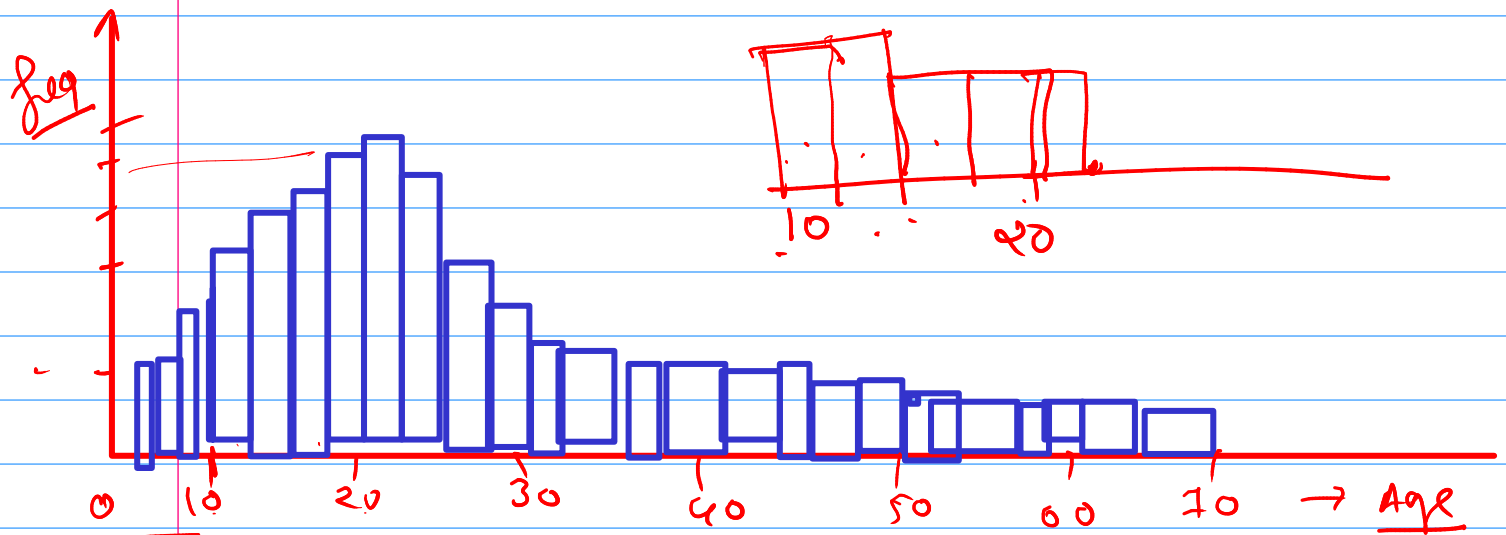


② Univariate (Numerical feature)

eg $\Rightarrow$ Age | Fare | Price



Distribution $\Rightarrow$ Histogram $\Rightarrow$ Numerical



2

Bivariate

- ① Cat - Num / Num - Cat
- ② Num - Num
- ③ Cat - Cat

① Cat & Cat

CrossTab
Contingency Table

200 Passengers data

Gender	Pclass
m	GN
	SL
	3AC
	1AC
	2AC
F	GN
	SL
	3AC
	1AC
	2AC

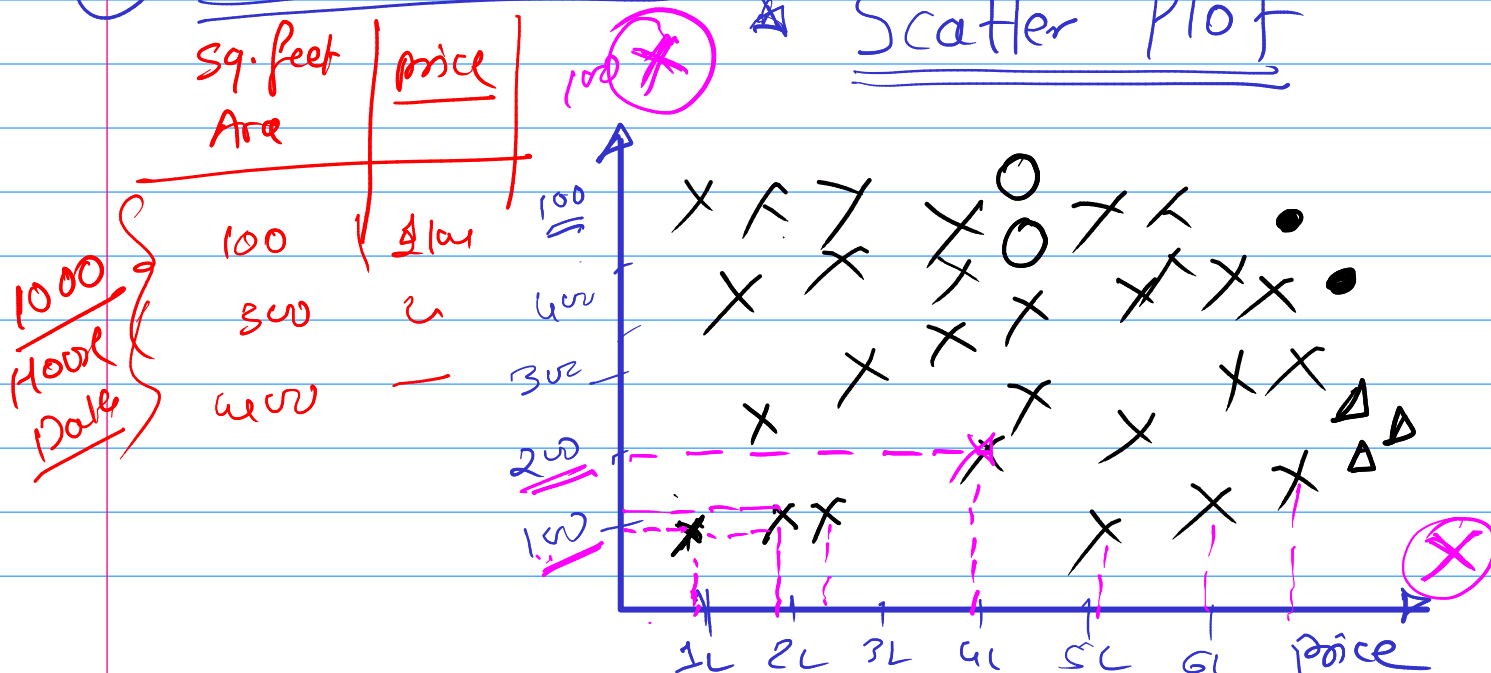
Plans

Gender	GN	SL	3AC	2AC	1AC
m	31	03	81	17	18
F	41	09	30	20	47

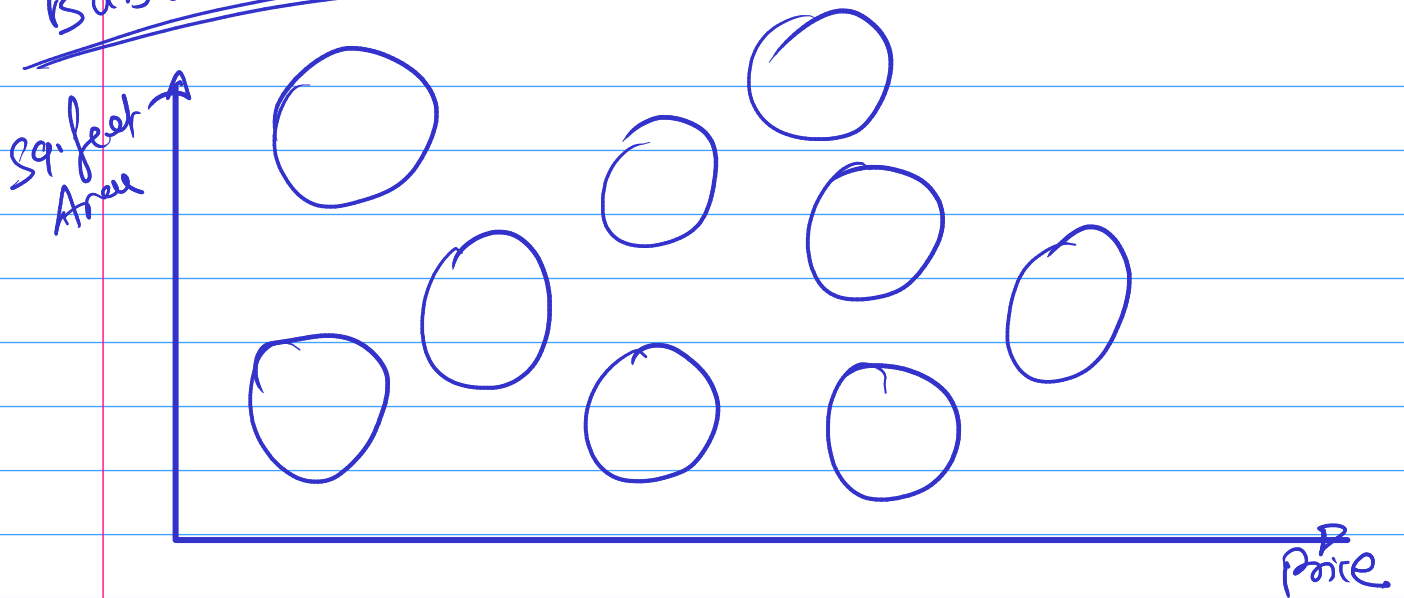
GN
17 m-2AC

② Num - Num

Scatter Plot



Bubble chart

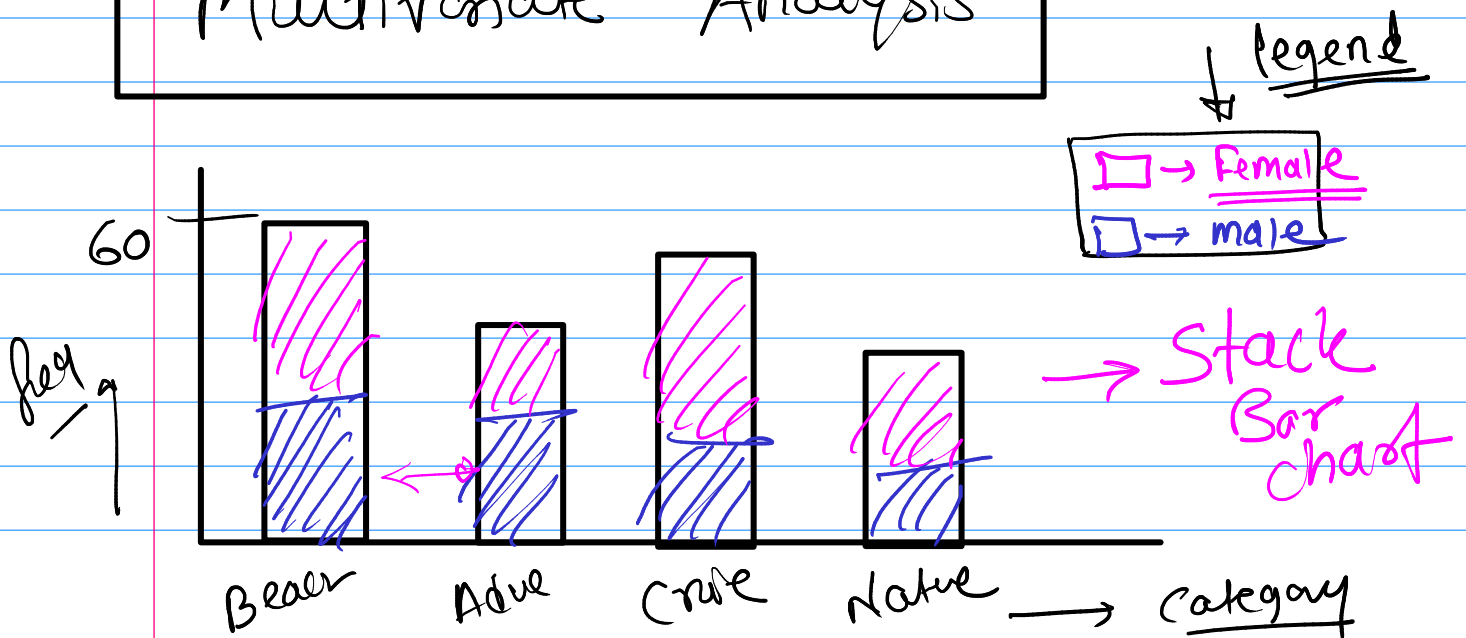


NUM - Cat / Cat - NUM

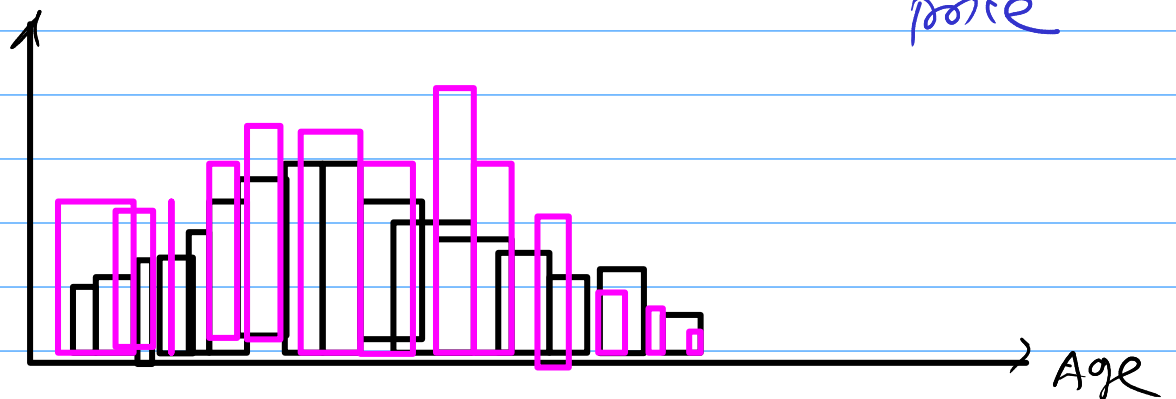
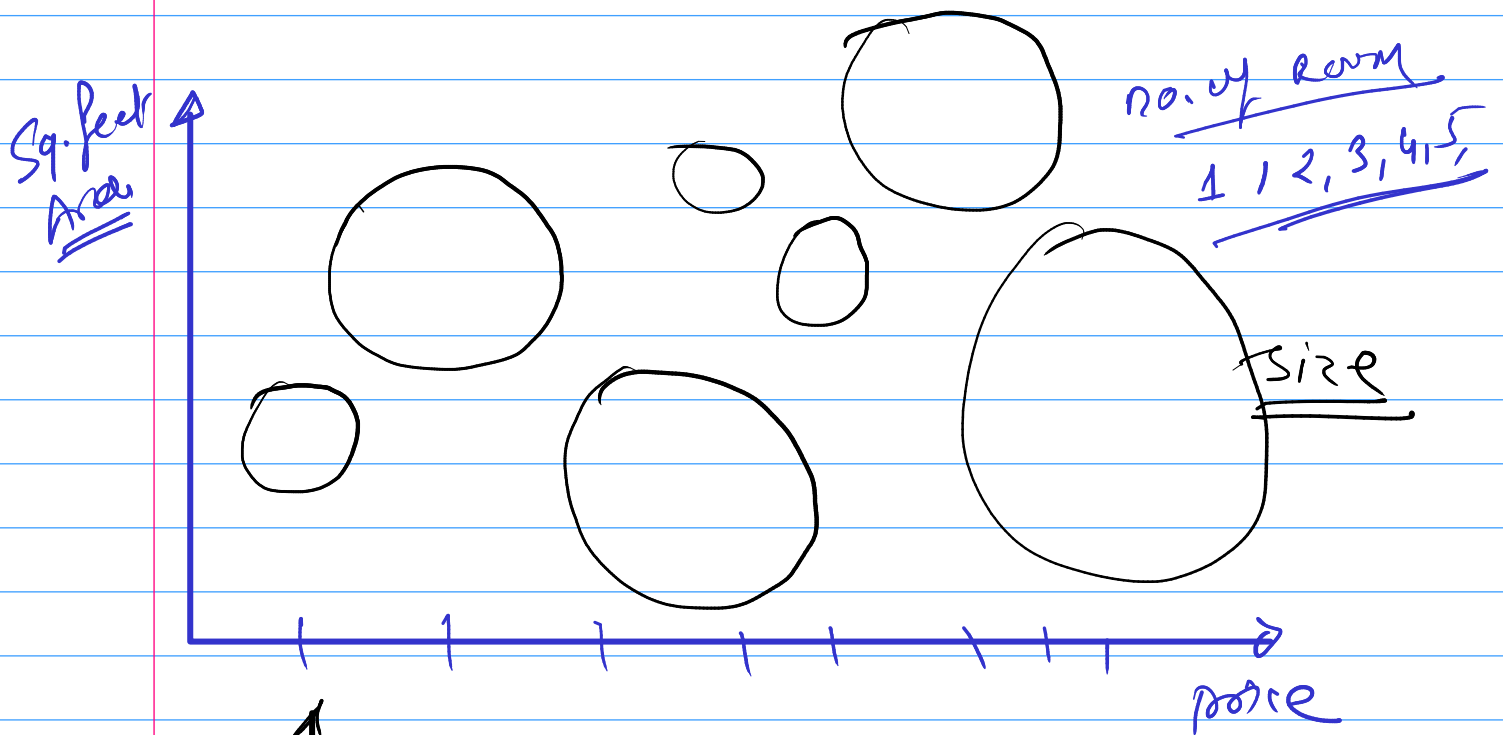
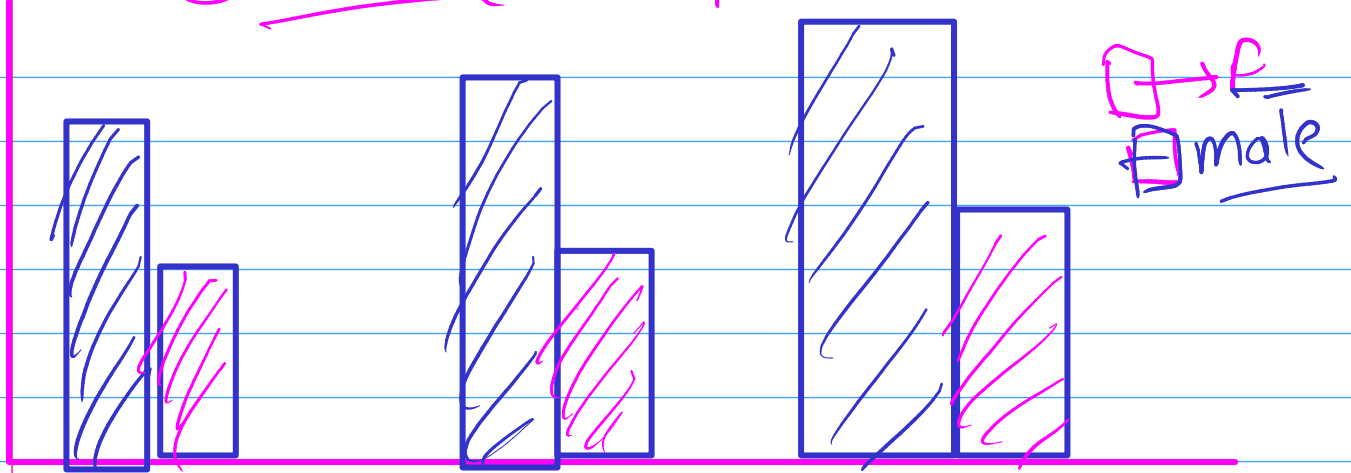
Gender	Count
male	30
female	45

→ Bar chart

Multivariate Analysis



Clustured Bar / Column chart





Time Series

o =

Hour, m, Day,
month, year,

Line chart

